

Canonical Modulus-Complete Heads for Projection-Free Spectral Factors

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Abstract

The rank-growing tail theorem of RH-282 requires an actual noisy spectral head, not a fitted root list. For any Hilbert–Schmidt operator and threshold $q > 0$, we define the head to contain every algebraic eigenvalue of modulus strictly larger than q . The head is finite, conjugation complete for real operators, and has cardinality at most the squared spectral mass divided by q^2 . It is the unique smallest spectral submultiset whose complement has spectral radius at most q . The strict threshold convention also resolves ties. Its genus-one product is an exact finite factor of $\det_2$, independent of eigenvector conditioning. Canonicity is relative to q and does not identify the selected roots with the deterministic monodromy shell.

1 Definition

Let $A \in \mathcal{S}_2(H)$ and let (μ_j) be its nonzero eigenvalues with algebraic multiplicity. For $q > 0$ set

$$H_q(A) = \{\mu_j : |\mu_j| > q\}, \quad T_q(A) = \{\mu_j : |\mu_j| \leq q\}.$$

The equality case belongs to the tail. This strict convention is part of the definition.

2 Minimal head theorem

Theorem 2.1. *Write $M = \sum_j |\mu_j|^2$. Then $H_q(A)$ is finite and*

$$\#H_q(A) \leq M/q^2 \leq \|A\|_2^2/q^2.$$

If a finite spectral submultiset F has the property that every eigenvalue outside F has modulus at most q , then $H_q(A) \subseteq F$. Thus $H_q(A)$ is the unique smallest admissible head.

Proof. Every member of H_q contributes more than q^2 to the convergent squared sum, proving finiteness and the count. If F omitted an eigenvalue with modulus greater than q , its complement would violate the radius condition. Hence every admissible F contains H_q . $\square$

Corollary 2.2 (real symmetry). *If A commutes with a real conjugation, then $H_q(A)$ is closed under complex conjugation, with equal algebraic multiplicities. Its finite canonical product has real Taylor coefficients.*

Proof. The algebraic spectrum of a real operator is conjugation invariant, and modulus is unchanged by conjugation. $\square$

3 Exact determinant split

The genus-one products

$$D_{H,q}(z) = \prod_{\mu \in H_q} (1 - z\mu)e^{z\mu}, \quad D_{T,q}(z) = \prod_{\mu \in T_q} (1 - z\mu)e^{z\mu}$$

are well defined, the first finite and the second locally normally convergent. RH-234 gives the exact identity

$$\det_2(I - zA) = D_{H,q}(z)D_{T,q}(z).$$

No invariant-subspace basis is needed.

Remark 3.1. The threshold q is an explicit design parameter. The theorem is canonical after q is fixed, not a claim that the dynamics chooses $q = 1/2$. It also does not prove that H_q equals the RH-272 counterloop atoms. Gates A–E remain false/open.